ELEC361 ANALOG COMMUNICATION SYSTEMS HW ASSIGNMENT **Fall 2023**

Submission Date: Nov. 10, 2023, Friday, in class, in the first hour.

E-postadan veya Teams üzerinden ödev kabul edilmeyecektir. *ω*'ya göre yapılan çözümler dikkate alınmayacaktır. Her türlü kopya işlemi hakkında yaptırım uygulanacaktır.

1) $x(t) = \cos(2\pi f_0 t), \quad y(t) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{t-kT_0}{\frac{T_0}{2}}\right)$

a) Find exponential Fourier series expansion of $x(t)$ and $y(t)$.
b) If $z(t) = x(t)y(t)$ and exponential Fourier series expansion coefficients of $x(t), y(t), z(t)$ are x_n, y_n, z_n respectively, find z_n by using $z_n = \sum_{l=-\infty}^{\infty} x_l\, y_{n-l}$.
c) Sketch $z(t)$ and find its exponential Fourier series expansion directly and compare it to the result in part b).

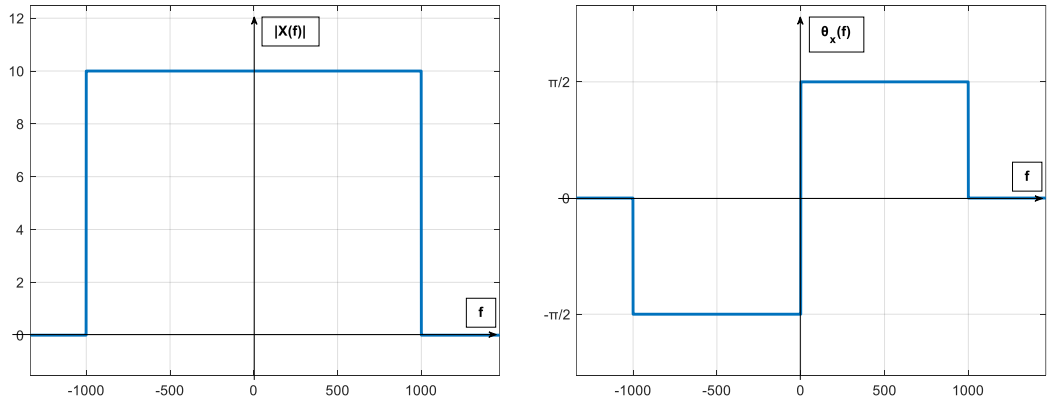
2)

a) If $X(f) = 10\Pi\left(\frac{f}{100}\right) * \Pi\left(\frac{f}{100}\right)$, find $X(f)$ by using convolution integral.
b) Find $x(t)$ by using inverse Fourier transform. Is $x(t)$ energy or power signal? Find its power or energy content.

3) If $x(t) = \begin{cases} cos(2\pi f_0 t) & , \quad -\frac{T_0}{2} \le t \le \frac{T_0}{2} \\ 0 & , \quad Else \end{cases}$, find $X(f)$ and sketch it.

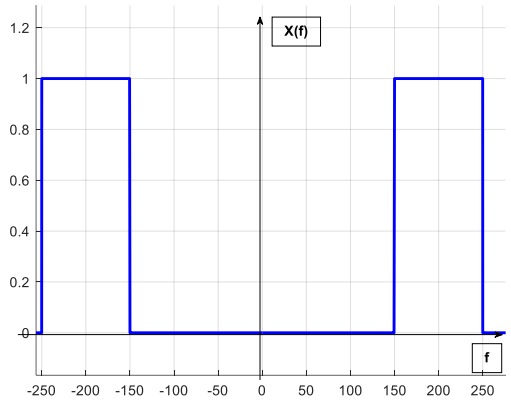
4) If $x(t) = \frac{1}{t}$, find $X(f)$ and sketch it. (Hint: Use duality property.)

5)



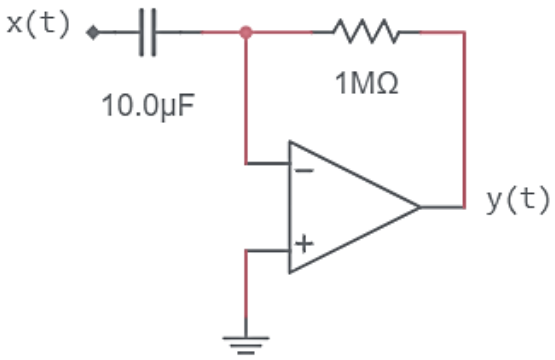
By using inverse Fourier transform, find $x(t)$.

6)



By using inverse Fourier transform, find $x(t)$.

7)



If the power spectral density (PSD) of the input voltage $x(t)$ is $S_x(f) = 10\Lambda\left(\frac{f}{100}\right)$, find the power of the output voltage $y(t)$.